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B.Tech. Degree VII Semester Examination November 2017**CS 1704 ANALYSIS AND DESIGN OF ALGORITHMS
(2012 Scheme)**

Time: 3 Hours

Maximum Marks: 100

PART A(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) State Masters theorem.
(b) Explain conquer technique with an example.
(c) Compare Insertion Sort and Quick Sort.
(d) Explain Binomial Heaps.
(e) Explain the various graph traversal methods.
(f) What are optimal binary search trees?
(g) Write notes on approximation algorithm.
(h) Explain graph colouring.

PART B

(4 × 15 = 60)

- II. Explain the different asymptotic notation used for specifying the growth rate of functions.

OR

- III. Explain the various criteria used for analyzing algorithms with suitable examples.

- IV. Explain heap sort algorithm. Derive the worst case and average case time complexity.

OR

- V. Explain Red black Tree. Show how insertion and deletion is performed.

- VI. Explain with an example strongly connected components of a directed graph.

OR

- VII. Explain Warshalls algorithm for finding transitive closure of a binary relation.

- VIII. Explain travelling salesman problem. Explain three possible strategies for TSP.

OR

- IX. What is bin packing problem? Explain how first fit decreasing strategy can be used for solving it.

